

Project 1

Image Registration using the [HPatches dataset](#)

- Use only the viewpoint sequences (v_*).
- Each sequence folder contains a reference image (1.ppm) and 5 target images (2.ppm to 6.ppm) captured from increasingly extreme viewing angles.
- Tasks per Sequence:
 - Estimate the homography matrix between the reference image and each of the 5 target images.
 - Compare your estimated matrix against the provided ground truth (H) by calculating the Corner Error. This is defined as the RMSE in pixel distance between the corners transformed by your estimated matrix and the ground-truth-transformed corners.
- Evaluation: Calculate and plot the average corner error across all processed sequences for each target image stage (2–6).

Project 2

Image Retrieval using the [Google Landmarks Dataset v2 Micro](#)

- There are images from 3103 distinct landmarks. In `gldv2_micro.csv`, you can find the global landmark ID for each image.
- The query images can be found in `val.csv` (and `train.csv`). Note that the `landmark_id` values in `val.csv` are different from the ones in `gldv2_micro.csv` for the same images. Therefore, you should cross-reference and map them accordingly via their unique filename.
- For computational efficiency, restrict your analysis to the images belonging to the first 100 classes in `val.csv` (`landmark_id` 0 to 99). Apply this filter to both for the query pool and the gallery database.
- For each filtered query image, retrieve the Top-5 most similar gallery images.
- Calculate the final Top-1 and Top-5 retrieval precision.

Project 3

Dogs vs. Cats

Project 4

Natural Scene Classification

Project 5

Real or Generated

Project 6

Leaf Classification

Project 7

Face Identification

Project 8

Medical Image Segmentation

Project 9

Car Detection

Project 10

Fruit Detection